



Anti-gastric adenocarcinoma activity of 2-Methoxy-1,4-naphthoquinone, an anti-*Helicobacter pylori* compound from *Impatiens balsamina* L. [☆]

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ABSTRACT

2-Methoxy-1,4-naphthoquinone (MeONQ) from *Impatiens balsamina* L. exhibited strong anti-*H. pylori* activity in our previous study. In this study, we investigated the cytotoxicity of MeONQ against gastric adenocarcinoma (MKN45 cell line) and propose the relevant mechanisms. MeONQ resulted in serious necrosis via superoxide anion catastrophe when the treatment doses were higher than 50 μ M, whereas apoptosis occurred at low treatment doses (25–50 μ M) through the caspase-dependent apoptosis pathway. Necrosis is the dominant mode of cell death. MeONQ exhibited high ability to induce gastric adenocarcinoma necrosis, showing good potential as a candidate agent for *H. pylori* infection related disease therapy.

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1. Introduction

Gastric cancer is the most common malignant tumor of the gastrointestinal tract, being strongly associated with *H. pylori* infection [1,2]. The prognosis of advanced gastric cancer is poor with a 5–15% five-year survival rate [3]. Induction of cell proliferation dysfunction and apoptosis are currently recognized as the actively chemotherapeutic strategies for carcinoma [4,5]. Furthermore, necrosis induction is already an effective treatment in early-phase clinical trials [6]. Cell proliferation is normally restrained through the cell cycle which is commonly divided into G1, S, G2, and M phases, and is regulated by the cyclin-dependent kinase/cyclin protein complex [7]. Some anti-cancer drug studies have focused on cell cycle phase arrest induction to prevent tumor progression [8].

Metabolic and therapeutic stresses induce acute intracellular NAD⁺ and ATP depletion accompanied by calcium and reactive

oxygen species (ROS) accumulation [6,9]. The calcium and ROS catastrophe provoke mitochondrial permeability transition (PT) pore opening through Bcl-2 related protein (e.g. death antagonists: Bcl-2, Bcl-XL; death agonists: Bax, Bak) regulations, and then the following oxidative burst results in widespread plasma membrane permeabilization and irreversible necrotic cell death [6,9,10]. On the other hand, the PT pore opening induces cytosolic substances [e.g. cytochrome c and apoptosis-inducing factor] to leak into cytosol thus switching the caspase-dependent pathway leading to apoptosis [5,6,9–11]. The roles of ROS productions, death receptors, PT, caspases, and Bcl-2 related proteins have been extensively studied in anti-cancer chemotherapy [4,5,9,11].

In our previous study, MeONQ (Fig. 1) was isolated from *I. balsamina* L. (Balsaminaceae) and exhibited strong anti-*H. pylori* activity being equivalent to that of amoxicillin [12]. Additionally, other bioactivities were also reported including anti-hepatocellular carcinoma [13], antipruritic [14], anti-inflammatory, antiallergic [15], and antifungal [16] activities.

The bioactivity of MeONQ is mostly due to its high redox potential [17]. MeONQ can be metabolized by flavoenzymes in living cells to form MeONQ-hydroquinone (MeONQH₂) which is quite unstable and possibly reacts with molecular oxygen to form semiquinone (MeONQ[•]) and superoxide anion. Under aerobic conditions, the semiquinone generates new MeONQ

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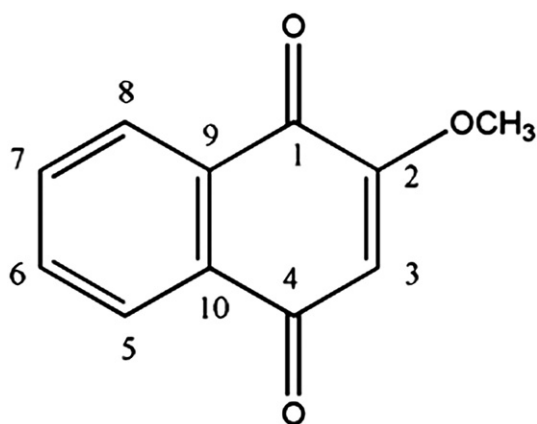


Fig. 1. Chemical structure of 2-Methoxy-1,4-naphthoquinone.

which reacts with the anion form of MeONQ-hydroquinone (MeONQH^-) to produce high amounts of ROS ($\text{O}_2^{\cdot-}$, $\text{MeONQ}^{\cdot-}$, H_2O_2 , $\text{OH}^\bullet$) resulting in organism injury or death [12,17,18].

In this study, we investigated the *in vitro* anti-adenocarcinoma activity of MeONQ and the relevant mechanisms. The time-course of MeONQ against adenocarcinoma cells, cell cycle distributions, morphological changes of cell nuclei, mitochondrial transmembrane potential ($\Delta\Psi_m$) disruption, and ROS generations as well as protein expressions of cytochrome c, pro-caspase-3, and Bcl-2 family are well defined herein.

2. Materials and methods

2.1. Cells

MKN45 cells (a human stomach cancer, poorly differentiated adenocarcinoma, JCRB0254) was obtained from Japanese Collection of Research Bioresources (JCRB; Osaka, Japan). MKN45 cells were cultivated in Rosewell Park Memorial Institute 1640 medium (RPMI 1640; Gibco, NY, USA) supplemented with 10% fetal bovine serum (FBS; Gibco, NY, USA) at 37 °C in an atmosphere of 5% CO_2 .

2.2. Cell viability assay

A methylthiazol-2-yl-2,5-diphenyl tetrazolium bromide (MTT) (Amresco, Ohio, USA) method was used for the determination of time/dose-dependent courses of 2-Methoxy-1,4-naphthoquinone (MeONQ; Sigma-Aldrich, MO, USA) against MKN45 cells [19]. Briefly, MKN45 cells were seeded onto a 96-well plate (1×10^6 cells/ml RPMI1640 medium, 0.1 ml) and cultivated at 37 °C in an atmosphere of 5% CO_2 for 24 h (80% confluence). The cell monolayers were washed with PBS and treated with 2-fold dilutions of MeONQ in RPMI1640 medium at 37 °C in an atmosphere of 5% CO_2 for 0–72 h. Non-MeONQ treated cells were used as the control. In the indicated periods of incubation, the cells were washed with PBS, 20 μl of 5 mg/ml MTT were added, and then incubated at room temperature (RT) for 2 h. Dimethylsulfoxide (DMSO; Tedia, Ohio, USA) was added, and the optical density (OD) was measured with a FLUOstar

OPTIMA spectrophotometer (BMG Labtech, Germany) at 570 nm. The relative cell viability (%) was calculated as the percentage of the ratio of OD_{570} of the treatments to that of the control. The 50% cytotoxic concentration (CC_{50}) was calculated by regression analysis of the dose–response curve of relative cell viability after 48 h incubation.

2.3. Cell cycle assay

MKN45 cell monolayers were incubated in the presence of 2-fold dilutions of MeONQ in RPMI1640 medium on a 24-well plate for 24 h. Non-MeONQ treated cells were used as the control. The cells were washed with PBS, lysed with trypsin-EDTA (Gibco, NY, USA), fixed with 95% ethanol for 12 h, and then centrifugated at $130 \times g$ for 3 min for the cell pellet collection. A volume of 1 ml of PBS containing propidium iodide (PI, 40 $\mu\text{g}/\text{ml}$) (Sigma-Aldrich, MO, USA) and RNase (0.1 mg/ml) (Sigma-Aldrich, MO, USA) was mixed with the cell pellets and incubated at 37 °C in the dark for 30 min. Cell cycle phases were measured by a flowcytometry (Cytomics FC500, Becton Coulter, USA) (Ex: 488 nm/Em: 620 nm) and phase distributions were calculated with Multicycle software (Phoenix Flow Systems, Inc. San Diego, USA). At least 10,000 cells were scanned. The experiment was performed in triplicate.

2.4. Nuclear morphology

A Hoechst 33258 (Sigma-Aldrich, Israel) fluorescent staining method was used. Hoechst 33258 can bind adenosine-thymidin-rich regions of DNA commonly used for labeling DNA in fluorescence microscopy and fluorescence-activated cell sorting [20]. MKN45 cell monolayers were incubated in the presence of 2-fold dilutions of MeONQ in RPMI1640 medium on a 24-well plate for 24 h. Non-MeONQ treated cells were used as the control. The cells were washed with PBS and fixed with 4% paraformaldehyde (Sigma-Aldrich, MO, USA) for 20 min. A volume of 0.2 ml of Hoechst 33258 (5 $\mu\text{g}/\text{ml}$) were added to each well and stood at RT for 5 min. Nuclear morphology of apoptotic cells with condensed nuclei was examined under a Zeiss LSM510 laser scanning confocal microscope image system with Zeiss 63X Plan-Apochromat objective (Germany).

2.5. YO-PRO-1/PI double staining

YO-PRO-1/PI double staining methods were used to distinguish from apoptotic and necrotic cells. The YO-PRO-1 dye (Molecular Probes, Eugene, OR, USA) can react with apoptotic cells' DNA or RNA in fluorescent green, whereas the fluorescent red derived from PI dye only occurs in necrotic cells. MKN45 cell monolayers were incubated in the presence of 2-fold dilutions of MeONQ in RPMI1640 medium on a 24-well plate for 24 h. Non-MeONQ treated cells were used as the control. The cells were washed with PBS, lysed with trypsin-EDTA, and centrifugated at $130 \times g$ for 3 min to collect the cell pellets. The cell pellets were incubated with YO-PRO-1 at 1 μM of final concentration at RT for 30 min; subsequently, PI at 1 $\mu\text{g}/\text{ml}$ of final concentration was added and incubated in the dark at RT for 5 min. The fluorescence derived from YO-PRO-1 and PI (Ex: 488 nm/Em: 525 and 620 nm, respectively) was detected by

a flowcytometry (Cytomics FC500, Bechman Coulter, USA). At least 10,000 cells were scanned. The experiment was performed in triplicate.

2.6. Mitochondrial transmembrane potential ($\Delta\Psi_m$) assay

JC-1 (5,5',6,6'-Tetrachloro-1,1',3,3' tetraethylbenzimidazoly carbocyanine iodide) (Molecular Probes, Eugene, OR, USA) is a cationic and lipophilic dye. In healthy cells, the dye stains the mitochondrial matrix and accumulates as aggregates in fluorescent red, whereas, in apoptotic and necrotic cells, JC-1 cannot accumulate in the mitochondria and remains in cytoplasm in a green fluorescent monomeric form [21]. $\Delta\Psi_m$ is expressed as the ratio of the red fluorescence intensity to that of the green one. The cell pellets described in the YO-PRO-1/PI staining section were suspended with PBS and stained with JC-1 (2 μ M of final concentration) in the dark at RT for 30 min. After centrifugation (130 $\times$ g, 3 min), the cell residues were suspended in PBS and the developed fluorescence measured at Ex 485 nm/Em 590 (red) and 520 nm (green) by a FLUOstar OPTIMA spectrophotometer. The experiment was performed in triplicate.

2.7. ROS time course assay

The MeONQ-induced ROS production in the MKN45 cells was determined according to Pauloina et al. [22]. MKN45 cell monolayers were incubated in the presence of 2-fold dilutions of MeONQ in RPMI1640 medium on a 96-well black plate for 0–24 h. In the periods of incubation, the cells in each well were stained with 0.2 ml of 1.58 μ g/ml dihydroethidium (HE, Sigma-Aldrich, MO, USA) and 0.2 ml of 1 μ g/ml dihydrodichlorofluorescein diacetate (H₂DCF-DA; Molecular Probes, Eugene, OR, USA) at RT in the dark for 30 min, respectively, for the superoxide anion and hydrogen peroxide determination. The fluorescence derived from HE and H₂DCF-DA was measured at Ex 485 nm/Em 570 nm and Ex 485 nm/Em 520 nm, respectively, by a FLUOstar OPTIMA spectrophotometer. The experiment was performed in triplicate.

2.8. Western blot assay

MKN45 cell monolayers were incubated in the presence of 2-fold dilutions of MeONQ in RPMI1640 medium on a 6-well plate (or a 6 cm-petri dish for cytochrome c determination) for 24 h. Non-MeONQ treated cells were used as the control. The cell pellets were collected according to the YO-PRO-1/PI staining methods, lysed with cell lysis buffer (Promega Biotech, USA) in an ice bath for 30 min, and then the supernatants were collected by centrifugation (130 $\times$ g, 3 min) as the cell lysates (or the cytosolic fraction). The residues were dissolved in mitochondrial extraction buffer mix (Biovision, USA) as the mitochondrial fraction. Both cell lysates (or cytosolic fraction) and mitochondrial fraction were subjected to protein quantification and Western blot assay.

The cell protein quantification was performed with Bradford reagent [0.1 mg/ml Coomassie Brilliant Blue G-250 (Amersco, USA) dissolved in 85% phosphoric acid (Wako, Japan)] followed by OD₅₇₀ detection with a FLUOstar OPTIMA spectrophotometer. The cell protein samples (50–70 μ g of total protein) were subjected to electrophoresis on 12% polyacrylamide

gels (Sigma-Aldrich, MO, USA) and then transferred to a polyvinylidene fluoride membrane (PerkinElmer Life Science, USA). After blocking with 5% nonfat milk in PBS, the membrane was respectively probed with primary antibodies [anti-Bcl-2 (R&D system, USA), anti-Bax (R&D system, USA), anti-pro-caspase-3 (R&D system, USA), and anti-cytochrome c (Biovision, USA)] at 1:1000 dilution, followed by addition of peroxidase-conjugated goat anti-mouse IgG (H+L) (Jackson ImmunoResearch Laboratories, PA, USA) secondary antibody. β -Actin (Abcam, UK) at 1:10,000 dilution was used for normalization. Protein bands were visualized with enhanced chemiluminescence reagent (PerkinElmer Life Sciences, USA) and exposed to Fuji Medical X-Ray films (Japan). Results were analyzed by Gel-Pro Analyzer software, version 4.0.00.001 (Media Cybernetics, USA). The experiment was performed in triplicate.

2.9. Statistical analysis

Data for cell viability, cell cycle distributions, YO-PRO-1/PI double staining, $\Delta\Psi_m$ disruption, and ROS time course were subjected to one-way analysis of variance (one-way ANOVA) with Dunnett's test by SPSS 12.0 software (SPSS Inc., USA) using least significant difference to identify significant differences from the controls ($P < 0.05$, $P < 0.01$).

3. Results

3.1. Cell viability

As shown in Fig. 2, the MKN45 cell viability was decreased with increasing treatment doses until 30 μ M treatment. There were no significant differences at 30–60 μ M treatments and 24–72 h incubation times ($P < 0.05$). IC₅₀ for MeONQ against MKN45 cells was 24.01 ± 0.38 μ M after 48 h incubation.

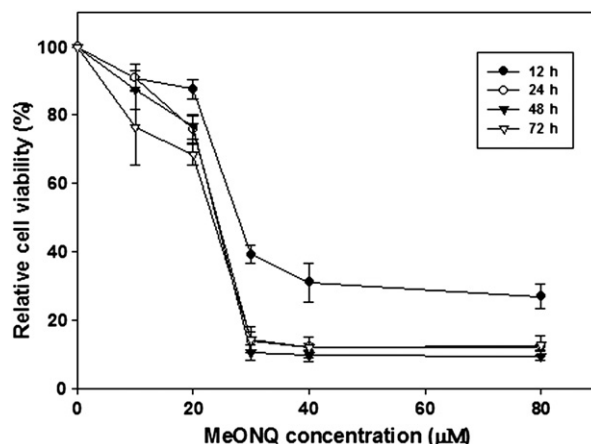


Fig. 2. Cell viability for MeONQ. MKN45 cells were treated with MeONQ (0, 10, 20, 30, 40, and 80 μ M) at 37 °C for 0–72 h. Data points presented are the mean $\pm$ standard deviation ($n = 3$).

3.2. Cell cycle distribution

As shown in Fig. 3a and b, 25–200 μ M MeONQ treatments induced G₀/G₁ phase rate decreases (79–89% of that of the control) and S and G₂/M phase rate increases (114–130 and

144–156% of that of the control, respectively). Furthermore, the subG₀/G₁ phase rates increased at 25 and 50 μ M treatments (360–491% of that of the control). Overall, MKN45 cells were treated with MeONQ resulting in apoptosis and the S and G₂/M phases of cell cycle arrest.

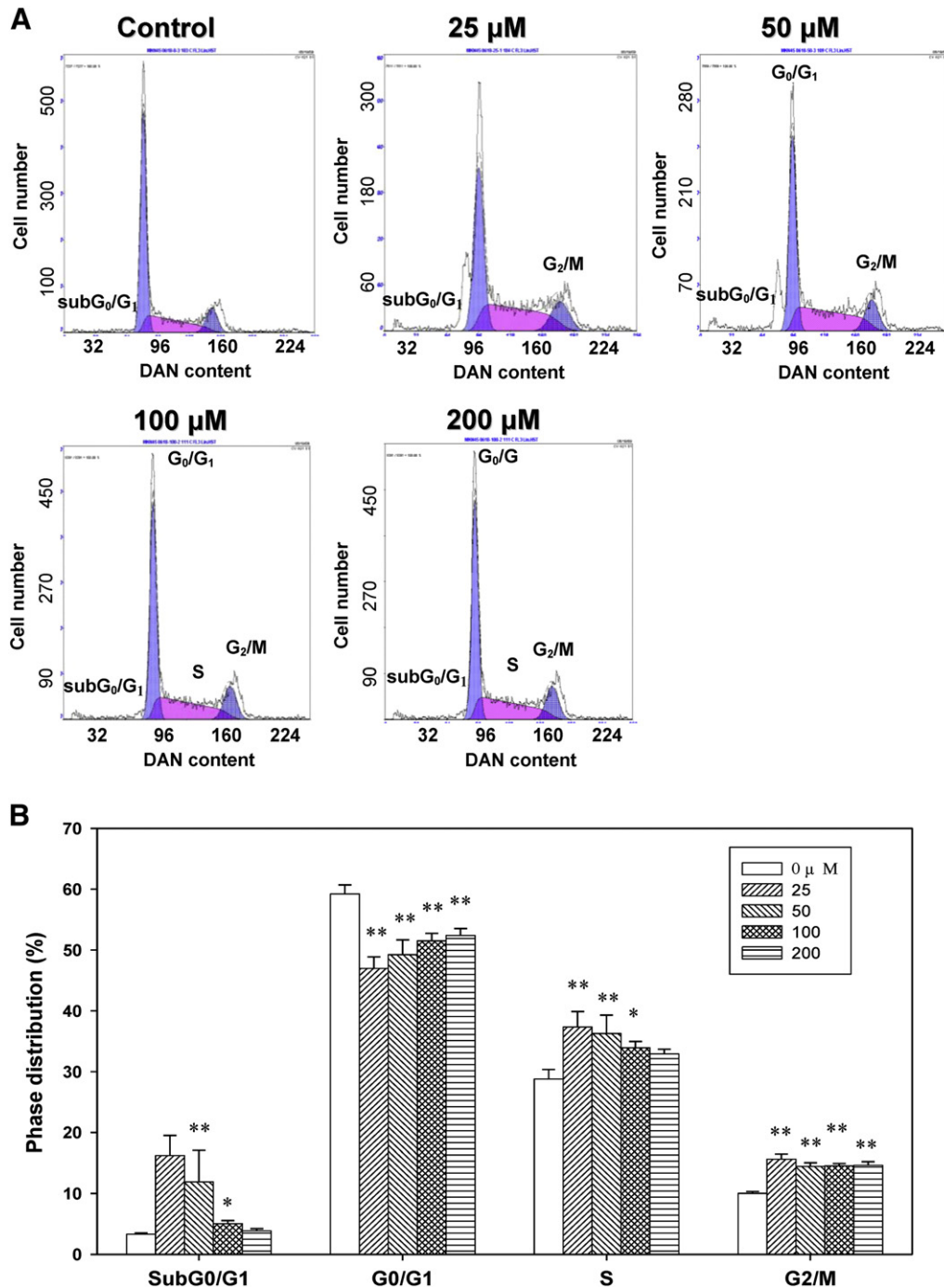


Fig. 3. Effect of MeONQ on the cell cycle distribution. MKN45 cells were treated with MeONQ (0, 25, 50, 100, and 200 μ M) at 37 °C for 24 h followed by flow cytometry analysis. Data presented are: A. histograms of the cell cycle distributions; and B. the relative distributions of each phase. Bars presented are the mean $\pm$ standard deviation ($n = 3$) with single and double asterisks showing the significant difference from the controls at $P < 0.05$ and $P < 0.01$, respectively.

3.3. Apoptotic and necrotic cell distributions

As shown in Fig. 4, the normal cell rates dramatically decreased and the late apoptotic/necrotic cell rates sharply increased with increasing treatment doses (12.5–200 μ M), for which 84.73 ± 2.22 – 93.77 ± 1.54 % (683–756% of that of the control) of late apoptotic/necrotic cell rates were measured at 50–200 μ M treatments after 24 h incubation.

3.4. Nuclear morphology

As shown in Fig. 5, fragmented and condensed chromatin DNA of cell nuclei, a significant characteristic of apoptotic cells, were observed in the 25 and 50 μ M MeONQ treated cells after 24 h incubation, with the rate for the 25 μ M treatment higher than that of the 50 μ M treatment. When the treatment doses were equal to or higher than 100 μ M,

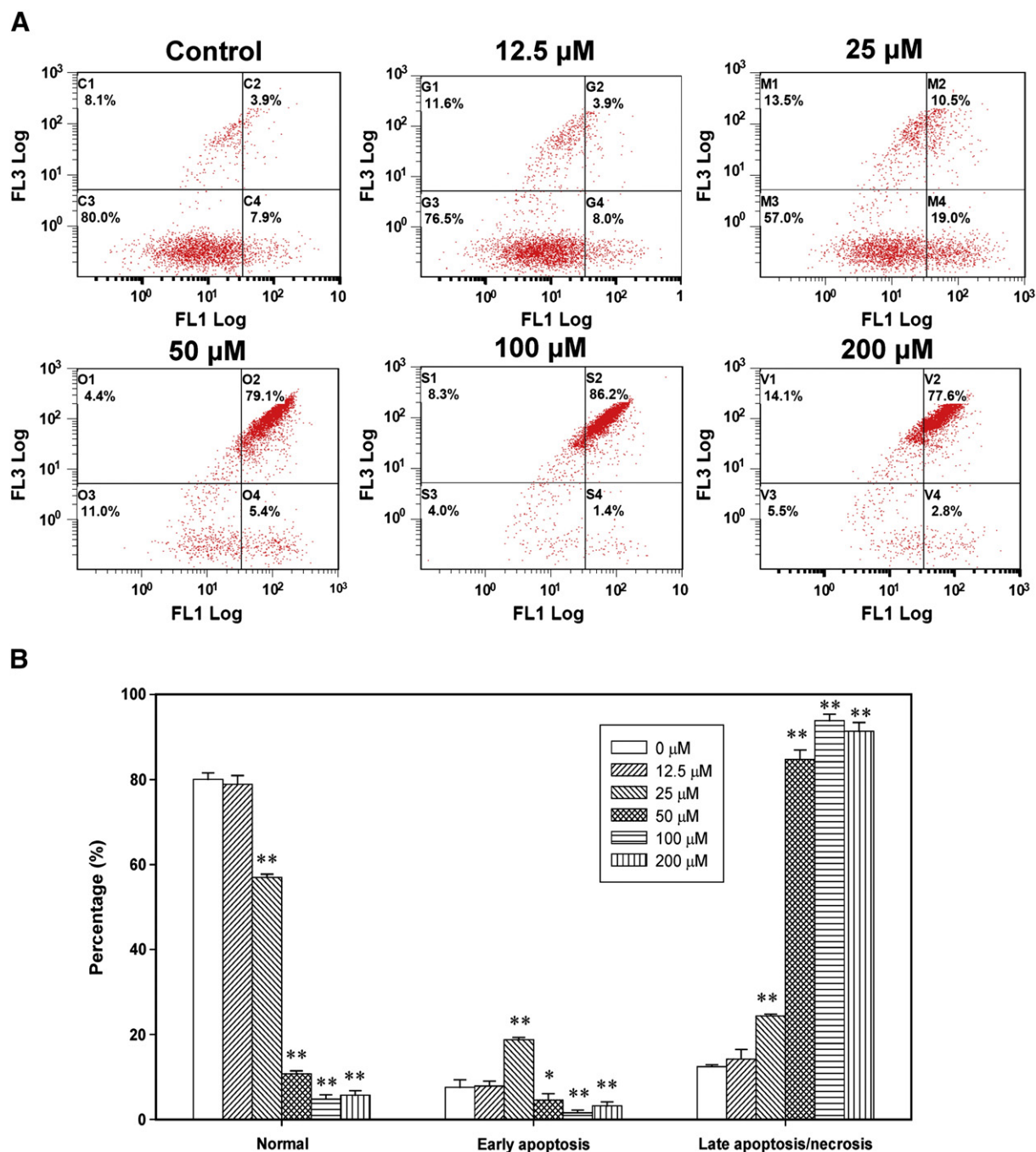


Fig. 4. Effect of MeONQ on the MKN45 cell damage. MKN45 cells were treated with MeONQ (0, 25, 50, 100, and 200 μ M) at 37 $^{\circ}$ C for 24 h, stained with YO-PRO-1/PI, and then subjected to flow cytometry analysis. Data presented are: A, histograms of cell damage; and B, the relative distributions of each stage. Bars presented are the mean $\pm$ standard deviation ($n=3$) with single and double asterisks showing the significant difference from the controls at $P<0.05$ and $P<0.01$, respectively.

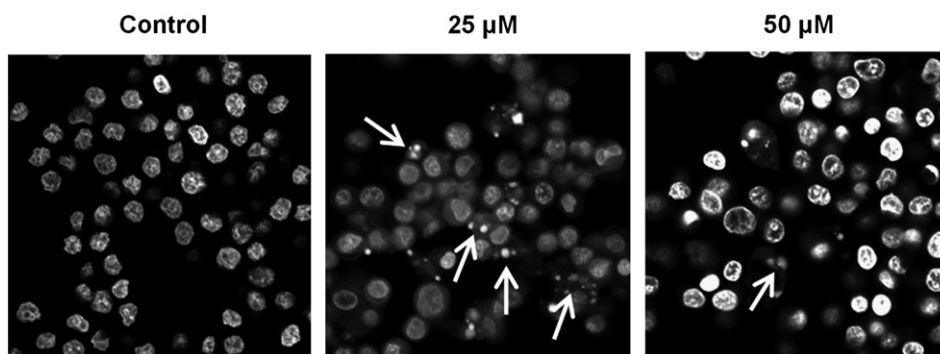


Fig. 5. Morphological changes of cell nuclei. MKN45 cells were treated with MeONQ (25 and 50 μM) at 37 $^{\circ}\text{C}$ for 24 h followed staining with Hoechst 33258. The arrows indicate the fragmented and condensed DNA of chromatin.

condensed chromatin DNA was scarcely detectable (data not shown).

3.5. $\Delta\Psi_m$ disruption

As shown in Fig. 6, $\Delta\Psi_m$ dramatically disrupted at 50–200 μM treatments (0.18 ± 0.02 – 0.64 ± 0.08 of $\Delta\Psi_m$, 11–39% of that of the control) ($P < 0.01$).

3.6. Time-course for ROS levels

As shown in Fig. 7a, relatively high superoxide anion levels were induced at 50–200 μM treatments in the period of 2–10 h incubation, peaking at 100 and 200 μM treatments for 6 h incubation (321.05 ± 2.03 – $336.26 \pm 17.47\%$ of that of the control); however, when the incubation time was over 8 h, those levels sharply decreased even below the control. The hydrogen peroxide levels slightly increased at 12.5 and 25 μM treatments (102.48 ± 1.73 – $140.99 \pm 4.09\%$ of that of the control) during 24 h incubation. However, when the treatment doses were equal to or higher than 50 μM , their levels were much lower than the control,

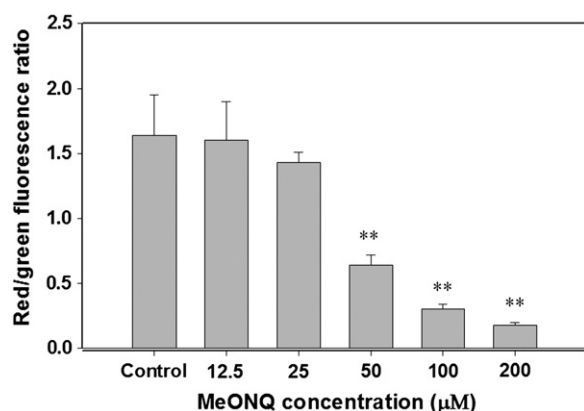


Fig. 6. Mitochondrial transmembrane potential ($\Delta\Psi_m$) disruption of MKN45 cells. MKN45 cells were treated with MeONQ (0, 12.5, 25, 50, 100, and 200 μM) at 37 $^{\circ}\text{C}$ for 24 h, stained with JC-1, and then red and green fluorescence were detected. Bars presented are the mean $\pm$ standard deviation ($n=3$) with double asterisks showing the significant difference from the controls at $P < 0.01$.

specifically for 10–24 h incubation, 3.29 ± 0.18 – $38.08 \pm 2.83\%$ of relative hydrogen peroxide levels being measured (Fig. 7b).

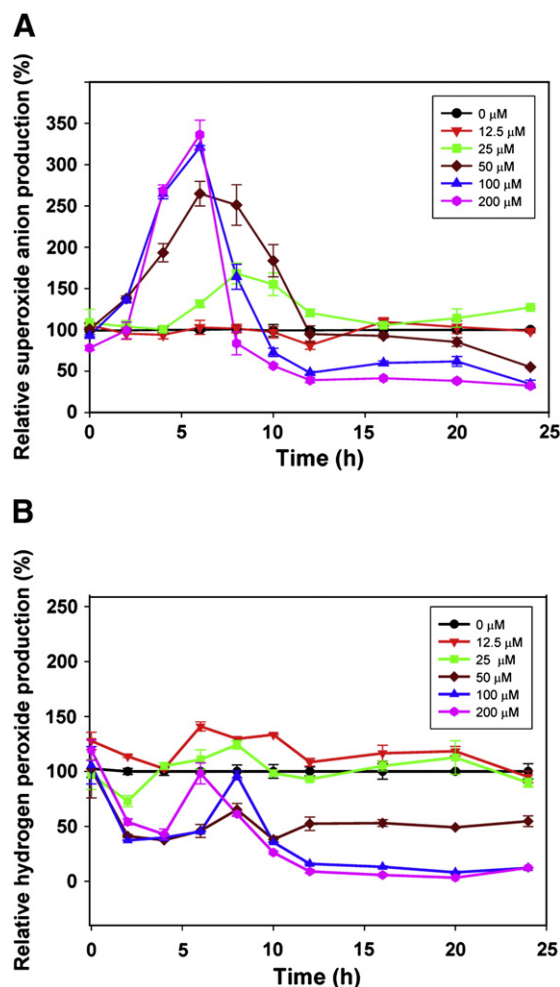


Fig. 7. Time course assay for the reactive oxygen species levels in MKN45 cells. MKN45 cells were treated with MeONQ (0, 12.5, 25, 50, 100, and 200 μM) at 37 $^{\circ}\text{C}$ for 0–24 h and then stained with A, hydroethidium and B, $\text{H}_2\text{DCF-DA}$ for the determination of superoxide anion (Ex 485/Em 570 nm) and hydrogen peroxide (Ex 485/Em 520 nm) levels, respectively. Data points presented are the mean $\pm$ standard deviation ($n=3$).

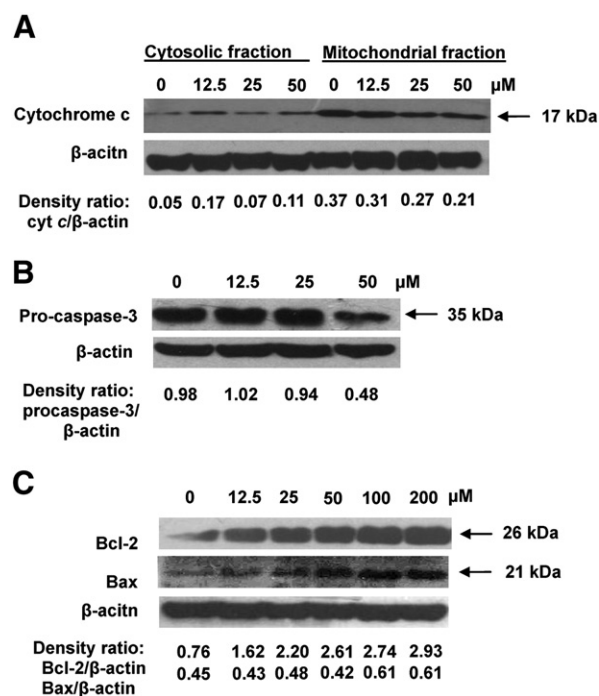


Fig. 8. Protein expressions of: A. cytochrome c; B. Bcl-2 family; and C. pro-caspase-3 in the MeONQ treated-MKN45 cells. MKN45 cells were treated with MeONQ at 37 °C for 24 h.

3.7. Protein expressions

As shown in Fig. 8, cytochrome c protein expressions increased in cytosol and decreased in mitochondria with

increasing treatment doses, of which density ratios were 220 and 57% of that of the controls at 50 μM treatment, respectively (Fig. 8a). Pro-caspase-3 protein expressions significantly decreased at 50 μM treatment, 49% of that of the control (Fig. 8b). Both Bcl-2 and Bax protein expressions significantly increased with increasing treatment doses; the density ratios were 386 and 136% of that of the controls at 200 μM treatment, respectively (Fig. 8c).

4. Discussion

MeONQ was isolated from *I. balsamina* L. and exhibited the strongest anti-*H. pylori* activity in natural products reported in our previous study [12]. In this study, MeONQ exhibited anti-gastric adenocarcinoma activity with 24.01 ± 0.38 μM (4.52 ± 0.07 μg/ml) of IC₅₀ (Fig. 2). If we classify anti-gastric cancer natural product activity as high [IC₅₀ < 10 μM (or < 1 μg/ml)] [19], moderate [IC₅₀: 10–50 μM (or 1–10 μg/ml)] [23], and low [IC₅₀: 50–500 μM (or 10–100 μg/ml)] [24]; the MeONQ activity was moderate but close to high.

In Fig. 3, MeONQ treatments caused MKN 45 cell apoptosis as well as arrested MKN45's cell cycle at the S and G2/M phases; however, both changes were not strong, indicating that apoptosis and cell cycle arrest were not the main routes of MeONQ against MKN45 cells. Similar results can also be found in Fig. 4, as apoptosis was only detected in 25 μM MeONQ treated cells; on the contrary, high populations of necrotic cells were observed at high-dose (50–200 μM) treatments along with high normal cell rate decreases. Aside from this evidence, more findings were further illustrated in Figs. 5 and 6. Fragmented and condensed chromatin DNA is a significant characteristic of apoptotic cells, but not of necrotic cells [6,9,10]; however, fragmented chromatin DNA was only measured at low-dose (25 and 50 μM) treatments (Fig. 5) in our study. Additionally, the cell density under microscopic

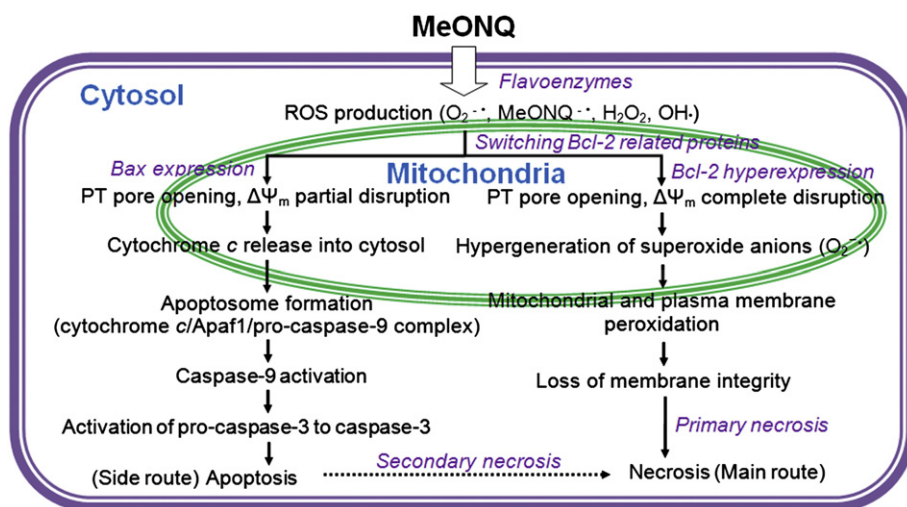


Fig. 9. Consequences of MKN 45 cell death induced by MeONQ. MeONQ enters into MKN45 cells and produces ROS ($O_2^{\bullet-}$, $MeONQ^{\bullet-}$, H_2O_2 , $OH^{\bullet}$) via flavoenzymes thus switching Bcl-2 related protein regulations. If there is Bcl-2 hyperexpression resulting in PT pores opening and $\Delta\Psi_m$ complete disruption, superoxide anion ($O_2^{\bullet-}$) catastrophe results. Plasma membrane is peroxidized by the hypergeneration of superoxide anions, losing its integrity and leading to necrosis (primary necrosis). On the other hand, Bax expression results in PT pores opening and $\Delta\Psi_m$ partial disruption, cytochrome c is released into the cytosol and forms an apoptosome with Apaf1 and pro-caspase-9. After caspase-9 activation, pro-caspase-3 is subsequently activated to produce caspase-3, which is then followed by the intrinsic caspase-dependent pathway leading to apoptosis. The apoptotic cells also can proceed to necrosis (secondary necrosis). Of the two types of cell death, necrosis is dominant.

examination decreased with increasing treatment doses (data not shown). The mitochondrial transmembrane potential results from the unequal distribution of ions on the inner mitochondrial membrane. When a PT pore opens, the $\Delta\Psi_m$ dissipation is found, which occurs in both apoptotic and necrotic cells with partial and complete $\Delta\Psi_m$ disruption, respectively [9,25]. As shown in Fig. 6, 82–87% of $\Delta\Psi_m$ ruptured at 100 and 200 μM treatments; in other words, the mitochondrial and plasma membrane almost collapsed in the cells which exhibited necrosis. Summarizing the results of Figs. 2–6, MeONQ resulted in serious necrosis when the treatment doses were equal to or higher than 50 μM , whereas apoptosis occurred at low treatment doses but at low rates.

In order to further understand the mechanisms of MeONQ against MKN45 cells, the time course of ROS production, caspase-3 activity, and protein expressions of procaspase-3, Bcl-2, Bax, and cytochrome *c* were investigated. In Fig. 7a, superoxide anions were measured for 0–2 h incubation while MKN45 cells reacted with MeONQ. Notably, relatively high amounts (336% of that of the control at the highest level) of superoxide anions were detected in the periods of 2–8 h incubation at 50–200 μM treatments. MeONQ has high redox potential, which can produce high amounts of ROS in living cells [12,17,18,]. Those ROS are an inducer of PT [9]. Therefore, we suppose that the original superoxide anions derived from MeONQ in MKN45 cells induced PT pore opening; subsequently, supergeneration of superoxide anions in mitochondria peroxidized intracellular and cellular membranes leading to permeabilization, and thus bursting necrosis (primary necrosis) [6,9,26].

In the intrinsic apoptosis pathway, cytochrome *c* is originally located in the outer and inner membrane spaces of mitochondria. When PT pores open (ROS is one of key effectors), it is released into the cytosol and forms an apoptosome with Apaf1 (an adaptor of apoptosome) and pro-caspase-9. Following caspase-9 activation, pro-caspase-3 is subsequently activated to produce caspase-3 leading to apoptosis [9,27,28]. In our study, cytochrome *c* protein expressions decreased in the mitochondria and increased in the cytosol (Fig. 8a), as well as cytosolic pro-caspase-3 protein expression decreases (Fig. 8b). This evidence clearly indicates that MeONQ resulted in apoptotic MKN45 cell death via the intrinsic caspase-dependent apoptosis pathway.

Bcl-2 related proteins are the key regulators of mitochondrial membrane PT pore [9]. In the cells, a high Bcl-2 expression level induces primary necrosis, whereas low level induces apoptosis; Bcl-2 down-regulates the apoptotic pathway, not the necrotic pathway [29]. In our study, both death antagonist Bcl-2 and death agonist Bax protein expressions increased with increasing treatment doses; specifically, the Bcl-2 increase was higher than that of Bax (Fig. 8c). Those results supported our aforementioned assumption that MeONQ induced MKN45 cells toward both apoptosis and necrosis via the Bcl-2 trigger; which one was dominant depended on the ROS suppression.

Summarizing this study, MeONQ exhibited good anti-gastric adenocarcinoma activity through necrosis and apoptosis, with necrosis being dominant. We propose the relevant mechanisms shown in Fig. 9. MeONQ enters into MKN45 cells and produces ROS ($\text{O}_2^{\cdot-}$, $\text{MeONQ}^{\cdot-}$, H_2O_2 , $\text{OH}^{\cdot}$) via flavoenzymes thus switching Bcl-2 related protein regulations. If there is Bcl-2 hyperexpression resulting in PT pores opening and $\Delta\Psi_m$ complete disruption,

superoxide anions catastrophe results. Plasma membrane is peroxidized by the hypergeneration of superoxide anions, losing its integrity and leading to necrosis (primary necrosis). On the other hand, Bax expression results in PT pores opening and $\Delta\Psi_m$ partial disruption, cytochrome *c* is released into the cytosol and forms an apoptosome with Apaf1 and pro-caspase-9. After caspase-9 activation, pro-caspase-3 is subsequently activated to produce caspase-3, which is then followed by the intrinsic caspase-dependent pathway leading to apoptosis. The apoptotic cells also can proceed to necrosis (secondary necrosis). Of the two types of cell death, necrosis is dominant. MeONQ not only possess strong bactericidal *H. pylori* activity, but also has good ability to induce gastric cancer necrotic cell death. MeONQ exhibits good potential to be a candidate agent for the *H. pylori* infection related disease therapy.

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